

Sign Language to Voice Generation

Project Overview

This project focuses on converting sign language gestures into spoken words using image processing and machine learning. The system captures hand gestures through a webcam, classifies them using a K-Nearest Neighbors (KNN) model, and converts the recognized gesture into voice output using a text-to-speech engine. Technologies used include OpenCV for image processing, NumPy for numerical operations, Scikit-learn for machine learning, and pyttsx3 for speech synthesis. The aim of the system is to reduce the communication gap between sign language users and people who do not understand sign language.

Directory Structure

```
.
├── dataset/
│   ├── gesture1/
│   └── gesture2/
├── capture_gesture.py
├── train_model.py
├── gesture_recognition.py
├── gesture_model.pkl
├── label_map.pkl
└── README.md
```

Key Features

Gesture Data Collection

- Captures gesture images using a webcam.
- Saves gesture images in separate folders.

Preprocessing

- Converts images to grayscale.
- Resizes images to 64×64 pixels.
- Adds salt-and-pepper noise to improve robustness.
- Converts images into flattened feature vectors.

Model

- Uses K-Nearest Neighbors (KNN) classifier.
- Trained using scikit-learn.

Real-Time Recognition

- Captures frames from webcam.
- Processes frames and predicts gestures.
- Displays detected gesture on screen.

Text-to-Speech Output

- Uses pyttsx3 library.
- Converts recognized gesture text into spoken words.

Usage Instructions

1. Install Dependencies

```
pip install opencv-python numpy scikit-learn pyttsx3 joblib
```

2. Capture Gesture Dataset

```
python capture_gesture.py
```

3. Train the Model

```
python train_model.py
```

4. Run Real-Time Gesture Recognition
`python gesture_recognition.py`

Results

- The trained KNN model achieved high accuracy during testing.
- The system successfully recognized gestures in real-time and generated voice output.